Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/07/2017
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling Detached Bridgman Crystal Growth		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	derby@umn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	612-625-8881
Organization Name:	University of Minnesota		
PI Address 1:	Department of Chemical Engineering and Materials Science		
PI Address 2:	151 Amundson Hall, 421 Washington Avenue, SE		
PI Web Page:			
City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/01/2016	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
Contact Email:	jan.r.rogers@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Volz, Martin Ph.D. (NASA Marshall Space Flight Center) Croell, Arne Ph.D. (NASA Marshall Space Flight Center)		
Grant/Contract No.:	NNX16AC37G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The original goal of this research project was to study detached Bridgman growth in both microgravity and terrestrial environments. The research was to develop and apply rigorous thermal-capillary models, employing finite-element methods and coupled with furnace-scale heat transfer models, to describe detached solidification. This work was to promote collaboration between research groups at the University of Minnesota (UMN) and the NASA Marshall Space Flight Center (MSFC). A research team at MSFC was to focus on experimental aspects of detached Bridgman growth, while UMN researchers were to develop and apply sophisticated mathematical models to analyze the process. Toward developing tasks to promote these original objectives, Principal Investigator (PI) Derby visited Dr. Martin Volz at the NASA Marshall Space Flight Center on November 29, 2016, to discuss past progress and future research directions. However, after discussions with Dr. Volz and others, a shift in research focus was deemed appropriate, and this project has been redirected to address continuing opportunities for advancing the understanding of the mechanisms that determine the dynamics of particle transport in silicon melts and the pushing or engulfment of these particles at a moving solid-liquid interface, specifically small silicon carbide and silicon nitride particles during the solidification of silicon. This work has and will continue to coordinate with the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft- und Raumfahrt) and involving the Fraunhofer IISB and the University of Freiburg. Dr. Martin Volz of Marshall Space Flight Center is also involved with this work on particle engulfment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This understanding is vital to improving the quality of solar-grade, multi-crystalline silicon grown by directional solidification processes. Understanding the dynamics of particle pushing and engulfment will enable process development to minimize their effects on grown material. This will promote lower cost and higher quality of solar-grade, multi-crystalline silicon and will advance progress in commercial solar cell production. In addition, particle engulfment is also important in other physical processes, such as fabrication of metal-matrix composites, separation processes, cryogenic preservation of biological materials, and frost heaving.
Task Progress:	Unfortunately, we were not able to attract a first-year graduate student to the new project on detached Bridgman crystal growth from the incoming class of Fall 2016. As explained above, after PI Derby visited Dr. Martin Volz at the NASA Marshall Space Flight Center on November 29, 2016, future research directions were discussed. Subsequently, PI Derby continued to work on the particle engulfment project, and Dr. Jeffrey H. Peterson was also assigned to this project as part of his post-doctoral research. An important advance for this project was the discovery of a new scaling law that described the dependence of the critical velocity for the engulfment of silicon carbide (SiC) particles by a silicon solidification front on the size of the particle. Whereas all prior analytical models had predicted that the critical velocity scaled as either the inverse of particle radius or the radius raised to the -4/3 power, our finite element model showed a dependence on the radius raised to the -5/3 power. This finding was significant, since for the first time in over a decade of research on SiC inclusions in silicon, our model was able to provide a quantitative correlation with experimental results, and furthermore allowed for the unambiguous identification of the underlying physical mechanisms that gave rise to the observed behavior of this system. In particular, we identified a significant and previously unascertained interaction between particle-induced interface deflection (originating from the thermal conductivity of the SiC particle being larger than that of the surrounding silicon liquid) and curvature-induced changes in melting temperature arising from the Gibbs-Thomson effect. For a particular range of particle sizes, the Gibbs-Thomson effect flattens the deflected solidification interface, thereby reducing drag on the particle and increasing its critical velocity for engulfment. We showed via numerical calculations and analytical reasoning that these effects gave rise to the new scaling of the critical velocity to particle radius.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Tao Y, Peterson JH, Reimann C, Friedrich J, Jauss T, Sorgenfrei T, Croell A, Derby JJ. "Analysis of silicon carbide and silicon nitride particle engulfment during multi-crystal silicon growth for photovoltaics." Presentation at the 21st American Conference on Crystal Growth and Epitaxy (ACCGE-21), Santa Fe, NM, July 30–August 4, 2017. 21st American Conference on Crystal Growth and Epitaxy (ACCGE-21), Santa Fe, NM, July 30–August 4, 2017. Conference proceedings distributed at the meeting. , Jul-2017
Abstracts for Journals and Proceedings	Friedrich J, Jauss T, Croell A, Sorgenfrei T, Reimann C, Derby JJ. "The ParSiWal experiments onboard TEXUS 51 and TEXUS 53: Particle incorporation during solidification of solar silicon." Presentation at the 7th International Symposium on Physical Sciences in Space (ISPS-7) and 25th European Low Gravity Research Association Biennial Symposium and General Assembly (ELGRA-25), Juan-les-Pins, France, October 2–6, 2017. 7th International Symposium on Physical Sciences in Space (ISPS-7) and 25th European Low Gravity Research Association Biennial Symposium and General Assembly (ELGRA-25), Juan-les-Pins, France, October 2–6, 2017. Conference proceedings distributed at the meeting. , Oct-2017
Abstracts for Journals and Proceedings	Derby JJ, Tao Y, Reimann C, Friedrich J, Jauss T, Sorgenfrei T, Croell A. "Analysis of the engulfment of a foreign particle by a solidification interface: A new scaling for flow and interfacial forces involving SiC particles during silicon crystal growth." Presentation at the AIChE Annual Meeting, Minneapolis, MN, October 29–November 3, 2017. Proceedings, AIChE Annual Meeting, Minneapolis, MN, October 29–November 3, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Derby JJ. "The synergy of modeling and novel experiments for melt crystal growth research." Invited presentation at the Symposium on Fundamentals of Crystal Growth, 21st American Conference on Crystal Growth and Epitaxy (ACCGE-21), Santa Fe, NM, July 30–August 4, 2017. Symposium on Fundamentals of Crystal Growth, 21st American Conference on Crystal Growth and Epitaxy (ACCGE-21), Santa Fe, NM, July 30–August 4, 2017. Conference proceeding distributed at the meeting. , Jul-2017

Articles in Peer-reviewed Journals	Derby JJ, Tao Y, Reimann C, Friedrich J, Jauss T, Sorgenfrei T, Croell A. "A quantitative model with new scaling for silicon carbide particle engulfment during silicon crystal growth." Journal of Crystal Growth. 2017 Apr;463:100–9. https://doi.org/10.1016/j.jcrysgro.2017.02.012 , Apr-2017
Articles in Peer-reviewed Journals	Tao Y, Sorgenfrei T, Jauss T, Croell A, Reimann C, Friedrich J, Derby JJ. "Particle engulfment dynamics under oscillating crystal growth conditions." Journal of Crystal Growth. 2017 Jun 15;468:24-7. https://doi.org/10.1016/j.jcrysgro.2016.10.049 , Jun-2017
Articles in Peer-reviewed Journals	Friedrich J, Reimann C, Jauss T, Croell A, Sorgenfrei T, Derby JJ, Tao Y. "Engulfment and pushing of Si3N4 and SiC particles during directional solidification of silicon under microgravity conditions." Journal of Crystal Growth. 2017 Oct 1; 475:33–8. https://doi.org/10.1016/j.jcrysgro.2017.05.036 , Oct-2017
Papers from Meeting Proceedings	Derby JJ. "The synergy of modeling and novel experiments for melt crystal growth research." Plenary lecture at the VIII International Scientific Colloquium, Modelling for Materials Processing, Riga, Latvia, September 21-22, 2017. Proceedings of the VIII International Scientific Colloquim "Modelling for Materials Processing", Riga, Latvia, 2017. p. 15-20. https://doi.org/10.22364/mmp2017.1 , Sep-2017